Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

3 (a)	A1	Award one mark for any of: • Conversion of plain text into cipher (1) • Converting information/data into a code (1) • Converting information/data into an unreadable format (1)	Accept alternative wording	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

3 (b)	B1	Award one mark for any of: • To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1)	Accept alternative similar wording. Do not accept hacking.	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

3 (c) (i)	C1 C2 C3 C4	Award 4 marks for a correct response: • CAL|OTTA|MUIN|PO • PO|MUIN|OTTA|CAL Award 1 mark each up to a maximum of 3 for: • 4 lines of text (1) • Zigzag arrangement of letters (1) • Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1)		(4)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract

3 (c) (ii)	C5 C6	Award 2 marks for a linked explanation such as: • Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial		(2)

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Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

		and error / easy to use brute force to crack (1)		

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 1 of 5 · find part a.iii in this mark scheme extract

6(a)(i)	Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1)		2

Question Number	Answer	Additional Guidance	Mark
6(a)(ii)	Any **two** from: • Printer (1) • Bank card (magnetic strip) reader/scanner (1) • Bank card contactless/NFC reader/scanner (1) • Cash input and giving change device (1)	Allow any hardware that relates to paying by cash for MP4	2

Question Number	Answer	Additional Guidance	Mark
6(a)(iii)	To prevent unauthorised people from reading / understanding the bank card information (1)	Do not award 'to keep it from being hacked / accessed'.	1

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 2 of 5 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for each of (maximum of 4): 1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1) 2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1) 3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1) 4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1) 5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1)	For each mark point: • the general feature is followed by exemplification • needs to be a comparison, which may be implicit	4

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 3 of 5 · find part a.iii in this mark scheme extract

Question Number	Answer	Mark
6(c)	Indicative content: Characteristics – Any of the following: • Implemented in software. • Could involve the use of neural networks / clustering / modelling • Algorithms are trained by exposure to data that has been validated / checked • Algorithms learn by identifying commonalities between the validated / checked data • Algorithms make predictions for new data based on prior learning • Algorithms may incorporate recursive self-improvement Uses – Any of the following: • Game playing (opponents) • Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions) • Image / Audio processing (recognising objects/patterns, face recognition) • Logistics (scheduling, order fulfilment) • Control systems (cars, manufacturing, weapons, navigation • Diagnostic systems (medical, mechanical, electrical) • Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants) • Robotics (dangerous situations, help aged or disabled) Ethical issues – Any of the following: • Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes • Is a computer to be trusted to make decisions (life-death)? • If a computer discovers something that humans can't prove, should it be accepted as truth? • How do we know that the answer produced is correct? • Will people be comfortable interacting with machines that are considered intelligent? • Some jobs may be at risk, especially those requiring human skills such as empathy or creativity.	6

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 4 of 5 · find part a.iii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 5 of 5 · find part a.iii in this mark scheme extract

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Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

3 (a)	A1	Award one mark for any of: • Conversion of plain text into cipher (1) • Converting information/data into a code (1) • Converting information/data into an unreadable format (1)	Accept alternative wording	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 2 of 6 · find part a in this mark scheme extract

3 (b)	B1	Award one mark for any of: • To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1)	Accept alternative similar wording. Do not accept hacking.	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 3 of 6 · find part a in this mark scheme extract

3 (c) (i)	C1 C2 C3 C4	Award 4 marks for a correct response: • CAL|OTTA|MUIN|PO • PO|MUIN|OTTA|CAL Award 1 mark each up to a maximum of 3 for: • 4 lines of text (1) • Zigzag arrangement of letters (1) • Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1)		(4)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 4 of 6 · find part a in this mark scheme extract

3 (c) (ii)	C5 C6	Award 2 marks for a linked explanation such as: • Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial		(2)

5 of 16

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

		and error / easy to use brute force to crack (1)		

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Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

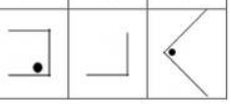
4 (a)	A1	**The only correct answer is B** A is not correct because this is decryption C is not correct because this is not an encryption D is not correct because this is not an encryption		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 6 · find part a in this mark scheme extract

4 (b)	B1 B2 B3	**Award 1 mark for each correct symbol** **Word** M A Y **Symbol** 		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 3 of 6 · find part a in this mark scheme extract

4 (c)(i)	C1	5		(1)

Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 4 of 6 · find part a in this mark scheme extract



Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

4 (c)(iii)	C3	Rail (Fence cipher)		(1)

Type MS · What encryption is and why we use it

Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · What encryption is and why we use it

Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A** is not correct because compression does not encrypt a file* ***B** is not correct because* encryption does not ensure both sides use the same key ***D** is not correct because* encryption does not stop hackers accessing a network		1

Type MS · What encryption is and why we use it

Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · What encryption is and why we use it

Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · What encryption is and why we use it

Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark